

Week 4 – Workshop Plan

Gravity: Density, Structures, and Non-Uniqueness

Geophysics 3A: Potential Fields & Geothermics

March 13, 2026

Duration: 3 hours

Setting: Workshop room (whiteboards; projector)

Purpose

This workshop develops intuition about how density varies in porous and compacted materials, how different subsurface structures relate to density contrasts, and how the buried sphere solution illustrates non-uniqueness in gravity. Students work from physical reasoning, analytic solutions, and qualitative cross-section analysis. The session prepares students for Week 5 (filters, upward/downward continuation, reduction-to-the-pole).

Learning Outcomes

- Estimate bulk density of porous materials using simple mixing laws.
- Explain how compaction and pore-fluid substitution change density.
- Predict density contrasts for various geologic structures.
- Interpret gravity anomaly width–depth relationships for the buried sphere.
- Recognise non-uniqueness and explain why lenticular shallow bodies can mimic deep spheres.

Roles and Grouping

Mixed groups of 4–5 with geology and physics/engineering backgrounds. Rotate roles: *Facilitator*, *Math Lead*, *Geology Lead*, *Skeptic/QC*, *Recorder*.

Session Flow (180 min)

0–10 min: Warm-up — “Why does density matter?”

In groups, list three factors controlling density of Earth materials. Share: how do porosity, mineral composition, and pore fluids alter density? (Students recall Week 2 density work with igneous samples.)

10–45 min: Activity 1 — Density of a Two-Component Porous Sandstone

Goal: Build intuition for porosity, compaction, and fluid-substitution effects.

Tasks (Worksheet A):

1. Consider a sandstone with quartz and feldspar fractions f_q and f_f and porosity ϕ . Estimate bulk density using:

$$\rho_{\text{bulk}} = (1 - \phi)(f_q \rho_q + f_f \rho_f) + \phi \rho_{\text{fluid}}.$$

2. Explore how reducing ϕ (compaction from burial) changes ρ_{bulk} .
3. Compute ρ_{bulk} for water-, air-, and brine-filled pore spaces.
4. Discuss: which fluid contrast would be measurable with gravity alone?

Groups record 3–4 bullet points about physical intuition (e.g., why air-filled rocks are anomalously light; why compaction increases density nonlinearly).

45–75 min: Activity 2 — Structure Density Reasoning Roundtable

Goal: Connect geology to density contrast and anomaly expectations.

Structures to analyse (Worksheet B):

- Buried sphere
- Buried cylinder
- Vertical dike
- Finite-extent sill
- Inclined dike
- Faulted contact
- Gently sloped basin

Group tasks:

1. For each assigned structure, sketch a simple cross-section.
2. Propose plausible density contrasts ($\Delta\rho$) given typical lithologies.
3. Predict qualitative anomaly shape: narrow/broad? symmetric/asymmetric? positive/negative?
4. One group presents each structure; whole class discusses similarities and differences.

75–105 min: Activity 3 — Buried Sphere: Width–Depth and Non-Uniqueness

Goal: Work with the buried sphere analytic solution and explore non-uniqueness.

Given: The vertical gravity from a buried sphere of radius a , density contrast $\Delta\rho$, and centre depth z_0 is:

$$g_z(x) = \frac{4\pi G}{3} \Delta\rho a^3 \frac{z_0}{(x^2 + z_0^2)^{3/2}}.$$

Tasks (Worksheet C):

1. Measure the *half-width* $x_{1/2}$ from provided plots and derive:

$$\frac{x_{1/2}}{z_0} = \sqrt{2^{2/3} - 1}.$$

2. Explain why increasing depth increases width but decreasing $\Delta\rho$ does not.
3. Discuss non-uniqueness: why can shallow, lens-shaped bodies (“lenticular”) mimic deep spheres?
4. Instructor demonstration: Python simulation showing buried sphere vs. lenticular bodies with same peak and width.

105–135 min: Activity D — Superposition: Build-an-Anomaly

Goal: Construct a plausible local cross-section and its gravity by summing components.

Provided: *Superposition Component Library* with scaled kernels (normalized shapes) for: basement step (sedimentary wedge), mafic dike (2D cylinder proxy), felsic pluton (broad polygon), thin regolith cap (negative slab).

Tasks:

1. Draft a geologically plausible cross-section (2–3 bodies) with $\Delta\rho$ from Activity A.
2. Place laterally, set depths/widths, scale amplitudes, and *sum* the responses (trace overlays on graph paper).
3. Annotate why the geology is plausible (lithology, structure, relative timing).

Deliverable: Labeled cross-section + composite gravity + legend of component contributions.

135–165 min: Activity E — Match-Three: Cross-Sections ↔ Gravity Profiles

Goal: Practice non-uniqueness reasoning with realistic models.

Provided: *Match-Three Handout* — three local-style cross-sections (A–C) and three gravity profiles (1–3) designed to be confusable.

Tasks:

1. Match each profile to the most plausible cross-section using: (i) half-width/FWHM (depth cue); (ii) edge slopes/asymmetry (contacts, wedges); (iii) amplitude scaling with $\Delta\rho$ and size.
2. Propose one additional observation to break any tie (e.g., rock density control, mapping a contact, short-station-spacing gradient, borehole, susceptibility).
3. Role inversion: *Geologists* present the math-based width–depth rationale; *Phys/Eng* defend the geology.

Deliverable: One page with matches (A→?, B→?, C→?) + tie-breaker justification.

165–180 min: Exit Ticket

1. Which factor (porosity, fluid, composition) most strongly affects density in shallow basins?
2. Which structural density contrast is hardest to distinguish with gravity alone?

Materials

- Worksheet A: Two-Component Sandstone Density
- Worksheet B: Structure Density Reasoning
- Worksheet C: Buried Sphere and Width–Depth
- Python code for instructor demo

Assessment

- Worksheet submissions (one per group).
- Quality of group discussion and presentations.
- Exit ticket reflection.